Case Study: Diagnosing & Resolving a ~$78K Billing Cliff

Direct-Billed Instant Quote (IQ) Policies | Revenue Operations Data Findings & Engineering Roadmap

Note: Proprietary company, partner, and employee names have been anonymized; metrics fuzzed for confidentiality.

1. Executive Summary

Across a 90-day cohort window, the company bound **~850 monthly-financed, allowlisted Instant Quote (IQ) policies** through direct-billed carriers. By tracking the earliest mortality ticket (PAYMENT_FAILURE or CANCELLATION_ALERT) per deal across this window, **~230 policies** had their first failure land specifically in a **Day 30–50 collision window**. This was identified as a structural billing-cliff pattern caused by corporate card down-payment mechanics, rather than sales training or customer lead quality.

- **~850 Monthly-Financed Binds** → ~230 policies hit the Day 30–50 cliff as their earliest mortality event.
- **Floor Loss Scenario:** ~100 policies received a CANCELLATION_ALERT in Days 30–50 → **~$31K in true revenue lost**.
- **Zero-Recovery Ceiling Scenario:** All ~230 cliff policies, if unrecovered → **~$78K in true revenue lost**.
- **Commission Rate Correction:** Corrected leadership's assumed commission rate from ~17% down to the true blended **~13.3%**.
- **Peak Single Day Failure:** Day 31, with **~60 deal failures** immediately following the carrier's automated rebill attempt.

2. The Core Problem & Mortality Timeline

The Bottleneck: PCI constraints forced sales agents to use corporate cards to front initial down payments for direct-billed IQ policies. Customers paid once via a one-time link, leaving no customer card on file with the carrier.
The Collision: Customers were never educated on this temporary financing. At Day 30, the carrier attempted to charge the corporate card for Month 2. The payment bounced, immediately triggering a Notice of Cancellation (NOC) sequence.
Operational Impact: Triggered ~700 weekly pending-payment tickets, an ~85% manual recovery failure rate, and severe strain on operations teams.

Stage	Timeline & Operational Reality
Day 1 — Bind	Agent collects down payment via one-time link and uses corporate card on carrier portal to bind. Customer is unaware of split transaction.
Days 1–29 — Void	Zero automated reminders go out prompting the customer to update payment method in carrier portal.
Day 30 — Collision	Carrier attempts to charge corporate card for Month 2. Rebill fails immediately.
Day 31+ — NOC	Carrier issues Notice of Cancellation. Ops receives ticket and begins manual triage (2x daily calls, bulk texts).
Day 45+ — Abandoned	Unresolved tickets (>2 weeks) are classified as lost causes and abandoned.

3. Query Methodology & Data Pipeline ('Cursor Logic')

To isolate true systemic mortality from noise, a custom SQL data pipeline was built using 'Cursor logic':
1. Financing Flag Proxy: The database `deals.financing` flag was broken (reading false on 100% of binds). A proxy metric was engineered: client_initial_payment < premium * 0.95, cleanly isolating the ~850 monthly-financed deals.
2. Deduplication & Fallback Join: Used DISTINCT ON (deal_id) ORDER BY timestamp ASC paired with a `company_id` fallback join to accurately attribute automated payment failure emails lacking deal IDs.
3. Financial Impact Formula: Calculated row-by-row lost ARR: premium * (commission_rate/100) * (11/12) based on exact deal terms, accounting for Month 1 earned commission.

4. Financial Impact Scenarios & Carrier Breakdown

Scenario	Deals	Lost Commission ARR	Fees Booked	Sunk CAC	True Revenue Lost
A: Floor (CA Ticket in Days 30–50)	~100	~$23.5K	~$20K	~$8K	**~$31K**
B: Ceiling (Any Ticket in Days 30–50)	~230	~$55.5K	~$53K	~$23K	**~$78K**

Carrier	Payment Failure (PF) Only	Cancellation Alert (CA) Only	Both (PF + CA)	Total Cliff Deals
Carrier A	~70	~30	~30	~130
Carrier B	~30	~10	~20	~60
Carrier C	~35	<5	~5	~40
Carrier D	0	0	~1	~1
Total	**~135**	**~45**	**~55**	**~230**

5. Sales Volume Concentration Analysis

Data analysis revealed heavy deal concentration: **Sales Rep A accounted for ~53% of floor deals (~50+ policies) and ~43% of ceiling deals (~100 policies)**. Further investigation confirmed this was driven strictly by automated lead-routing volume concentrated on top reps, rather than an isolated rep performance failure—preventing misattributed rep clawbacks.

6. Master Solutions Roadmap & Process Engineering

Immediate Solution (Deployed / In Trial): The COI Email Override

- *Mechanism:* Layered an automated IF/THEN filter directly onto the Operations team's existing IQ_COI_SEND email trigger—requiring zero new infrastructure.
- *Logic:* IF carrier IN (Target Allowlist) AND client_initial_payment < premium * 0.95, then override standard congratulatory copy with an urgent notice requiring portal payment method updates within 48 hours.
- *Impact:* Eliminates manual phone outreach and sets explicit Day-1 customer billing expectations.

Short-Term Roadmap: Automated Day-20 API Payment Check

- Automated cron job running 20 days post-bind to query carrier APIs, compare stored card last-4 digits against corporate card records, and trigger automated SMS alerts before Month-2 rebill attempts.

Long-Term Architecture: Full PCI Tokenization Vault & Forward Proxy

- Implement a 3rd-party PCI vault (e.g. VGS / Basis Theory) with forward proxy capabilities to securely collect customer card tokens at checkout and inject them into carrier binding APIs, permanently eliminating corporate card down-payments.

Non-Viable / Discarded Solutions: Rejected manual PDF payment auth forms (PCI violation), unsegmented email blasts (confuses active users), rep commission clawbacks (punitive for systemic gaps), and expanding manual offshore call queues (~15% recovery rate proved unscalable).